

[NJC/2025/Q2]

2 Name your Jupyter Notebook as `Task2_<your name>.ipynb`

A military special forces unit is organised into squads, each consisting of members with **unique** roles. Every year, new candidates apply to join the unit. A selection test is conducted to select suitable candidates. The file `candidates.txt` contains the scores of the selection test for these candidates.

Each line in the file has three fields in this order: **name**, **role**, and **score**. The fields are separated by a comma (,). Each candidate has a pre-assigned role based on their previous experience and training.

Your task is to write a program to select new candidates to form new squads in the special forces unit as well as to determine those that are not selected. You are allowed to use Python built-in functions in this task.

For each sub-task, add a comment statement, at the beginning of the code using the hash symbol "#", to indicate the sub-task the program code belongs to, for example:

```
In [1]: #Task 2.1
        Program code
Output:
```

Task 2.1

Write a Python function, `read_data(filename)` to read in the data from a text file and returns a list of tuples. Each tuple represents a candidate record in the file. The function needs to:

- take the name of a file `filename` as input.
- reads the contents of the file. Each line of the file has three fields: name, role, and score, separated by ','.
- returns a Python list of tuples, each tuple should be in the form

`(name, role, score)`

[3]

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Task 2.2

Write a Python function, `get_roles(candidates)`, where `candidates` is the list of tuples obtained in Task 2.1, and returns a Python **dictionary** containing each unique role as keys and a list of candidates who belonged to that role. Your code should dynamically determine the unique roles present in the `candidates` list and must not create the dictionary by hardcoding specific role names like 'Sniper' or 'Commander'.

Example: calling `get_roles(candidates)` should return

```
{'Sniper': [('Chadwick Griffin', 'Sniper', 95),
            ('Joella Wessner', 'Sniper', 93),
            :
            ('Luanne Lett', 'Sniper', 61)],
 'Commander': [('Marcella Daigneault', 'Commander', 95),
               :
               ('Ione Wolfe', 'Commander', 90)],
 :
 '2IC': [('Maryam Brosius', '2IC', 92),
         :
         ('Sharie Schall', '2IC', 58)]
}
```

[3]

Task 2.3

Using the data file `candidates.txt` and the functions `read_data()` and `get_roles()` implemented in Task 2.1 and Task 2.2 respectively, write Python code to :

- count the number of candidates in each unique role.
- determine the lowest and highest score obtained by the candidates in each unique role.
- output the statistics in the following format:

Roles	Count	Low Score	High Score
Sniper	8	61	95
:	:	:	:
:	:	:	:
2IC	7	58	92

[3]

[NJC/2025/Q2 Cont']

The selection criteria for the special forces unit to form new squads is as follows:

- A squad must contain members with unique roles, meaning no duplicate roles are allowed within the squad.
- The number of members in a squad is determined by the number of unique roles determined in Task 2.2, meaning if there are seven unique roles identified, all the squads formed must have **exactly** seven members. You cannot hardcode the number of members in the squad, this number must be dynamically determined.
- When selecting a candidate for a particular role in the squad, the candidate with **the highest score** in that role will be chosen first.
- A selected candidate can only be a member of one squad.

Task 2.4

Write a Python function, `form_squads(roles)` to form new squads based on the selection criteria described above. The function will:

- have an input, `roles`, a dictionary object obtained by calling `get_roles()` implemented in Task 2.2.
- returns a list of squads. Each squad is a list of candidates.

Example: calling `form_squads(roles)` should return

```
[('Chadwick Griffin', 'Sniper', 95),
 ('Marcella Daigneault', 'Commander', 95),
 ('Marg Thorington', 'Medic', 84),
 ('Terrence Shannon', 'Breach', 94),
 ('Shaunda Sieg', 'Navigator', 84),
 ('Reiko Stack', 'Communication', 83),
 ('Maryam Brosius', '2IC', 92)
],
[('Joella Wessner', 'Sniper', 93),
 ('Fredricka Gormley', 'Commander', 92),
 :
 ('So Atkins', 'Communication', 75),
 ('Russell Gillison', '2IC', 91)
],
:
[('Jalisa Stoudemire', 'Sniper', 76),
 ('Johnna Lecuyer', 'Commander', 48),
 :
 ('Marguerita Mciver', 'Communication', 48),
 ('Sharie Schall', '2IC', 58)
]
]
```

[NJC/2025/Q2 Cont ']

Task 2.5

Using the data file `candidates.txt` and the functions `read_data()`, `get_roles()` and `form_squads()` implemented in Task 2.1, Task 2.2 and Task 2.4 respectively, write Python code to output the new squads formed as follows:

- each squad is assigned a unique integer identifier.
- the average score of the squad's members.
- list of squad members with their names and roles.

Example:

```
Squad 1, Avg.Score(89.57)
(Chadwick Griffin,Sniper)
(Marcella Daigneault,Commander)
(Marg Thorington,Medic)
(Terrence Shannon,Breacher)
(Shaunda Sieg,Navigator)
(Reiko Stack,Communication)
(Maryam Brosius,2IC)
```

```
Squad 2, Avg.Score(85.29)
(Joella Wessner,Sniper)
(Fredricka Gormley,Commander)
(Hillary Curl,Medic)
(Lanita Sciortino,Breacher)
(Teodoro Negrin,Navigator)
(So Atkins,Communication)
(Russell Gillison,2IC)
```

```
:
```

```
Squad 7, Avg.Score(54.57)
(Jalisa Stoudemire,Sniper)
(Johnna Lecuyer,Commander)
(Virgilio Britt,Medic)
(Juli Barnhill,Breacher)
(Tashia Bowen,Navigator)
(Marguerita Mciver,Communication)
(Sharie Schall,2IC)
```

[3]

Task 2.6

Write Python code to determine and output the candidate/s who cannot be placed in any special forces squad and therefore was/were not selected. No output is needed if all the candidates were selected and placed in the squads.

Example:

Luanne Lett, who is a Sniper with a score of 61 is not selected [2]

Save your Jupyter Notebook for Task 2.